

Identity-Constrained Gamma Approximations for Stable Variational Inference

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ABSTRACT

The Gamma function is a function that appears throughout statistics, particularly in Bayesian inference over Gamma, Beta, and Dirichlet distributions. Modern Bayesian algorithms such as variational inference repeatedly evaluate the logarithm of the Gamma function and its derivatives, making both approximation accuracy and derivative quality important considerations for computational efficiency. This paper introduces a framework for constructing identity-constrained approximations of special functions. In contrast to fitting an approximation through pointwise error minimization, the proposed framework incorporates functional identities directly into the optimization objective. These identities serve as structural constraints that enforce global properties of the target function beyond local accuracy. The framework is demonstrated using the logarithm of the Gamma function as a case study through Bohr-Mollerup characterization, Euler recurrence relation, and Legendre duplication identity. Three optimization objectives are investigated: least-squares fitting, duplication-identity fitting, and a composite identity-constrained objective that balances approximation accuracy with structural identity preservation. A second phase evaluates these approximations in Bayesian inference using a Dirichlet-Multinomial variational inference benchmark. Because the Dirichlet posterior parameters can be computed exactly, every approximation is evaluated at the exact posterior parameters, isolating approximation error from optimization error. Performance is measured using Evidence Lower Bound (ELBO) accuracy together with gradient accuracy, evaluated through direct access to Log-Gamma, Digamma, and Trigamma functions. Experimental results demonstrate that identity-constrained approximations consistently provide the strongest performance within variational inference. In contrast, least-squares approximations generally achieve superior accuracy over a bounded grid of points. These findings suggest that preserving functional identities can improve downstream optimization performance even when traditional approximation error is not minimized among available models. The proposed framework extends to transformed Gamma functions, multivariate Gamma-related functions, and other special functions characterized by functional identities, providing a general methodology for constructing structurally informed approximation for use in computational statistics.

Keywords: Bayesian Optimization, Constraint Regression, Structure-Preserving Approximation

RESEARCH QUESTION

Can mathematical structure, rather than purely empirical approximation error, be used to construct smooth approximations to $(\log(\Gamma(x)))$ and its derivatives that improve downstream variational inference?

MOTIVATION

Gamma and Digamma functions occur throughout Bayesian computation, including Dirichlet variational inference. In many computational settings, approximation quality is evaluated primarily through pointwise error.

I instead became interested in a structural question: **Can known mathematical identities be built directly into the approximation family so that useful properties are preserved before fitting begins?** The goal is not simply to approximate a target function, but instead deems the target function latent to construct an approximation whose form respects the structure of the underlying special function dynamics.

APPROACH

I construct approximations to $(\log(\Gamma(x)))$ over a restricted interval using several structural components. The approximation family combines:

- polynomial baselines,
- residual correction terms,
- recurrence relations,
- reflection identities,
- analytically derived derivatives,
- and constrained coefficient fitting.

The fitting problem is performed numerically using SciPy differential evolution optimization. The resulting coefficients are frozen before downstream experiments. The approximation is then evaluated in a **Dirichlet variational inference** setting, where Gamma/Digamma evaluations affect optimization through the variational objective and its gradients.

EXPERIMENTAL DESIGN

The main experiment uses a full factorial design varying:

- baseline approximation,
- residual correction,
- fitting loss construction,
- derivative construction,
- and variational inference settings.

The primary downstream measurements include approximation behaviour and **gradient stability / gradient norm** during variational optimization.

INITIAL FINDING

Rather than selecting the approximation solely according to pointwise approximation error, the experiments evaluate whether the structural choices made during approximation construction propagate into **downstream optimization behaviour**.

WHY THIS MATTERS

The broader idea is to treat approximation design as a **structured inference problem**. As opposed to: fit function $\rightarrow$ assume downstream computation behaves well, the approach taken is: mathematical identities $\rightarrow$ constrained approximation family $\rightarrow$ optimization $\rightarrow$ downstream statistical behaviour. This suggests a general strategy for numerical Bayesian computation especially where access to the target function may be restricted or unavailable: useful mathematical structure can potentially act as an inductive bias for approximation, bypassing the need to evaluate the target function anywhere during fitting. This technique may appear especially useful for special functions that lack an explicit expression but can be uniquely characterized by a series of structural properties.

Fitting Method	Baseline	Residual	Phase 2 Ranking	Phase 1 Ranking	Change
Identity-Constrained	+	+	1	6	+5
Least Squares	+	+	2	4	+2
Least Squares	+	-	3	3	-
Identity-Constrained	-	+	4	5	+1
Identity-Constrained	+	-	5	7	+2
Least Squares	-	+	6	1	-5
Least Squares	-	-	7	2	-5
Identity-Constrained	-	-	8	8	-
Duplication-Constrained	+	+	9	9	-
Duplication-Constrained	-	+	10	10	-
Duplication-Constrained	+	-	11	11	-
Duplication-Constrained	-	-	12	12	-

Table 1. Approximation methods ranked on pointwise accuracy over a grid (phase one) and Bayesian variational inference performance (phase two).

FOLLOW-UP QUESTIONS

1. What sets of mathematical properties uniquely characterize a function?
2. How easily can the approximation framework be extended to other special functions?
3. How does identity-constrained fitting compare with unconstrained high-capacity approximators?
4. Can approximation error and downstream optimization behaviour be optimized jointly?
5. Can the same framework be applied to construct approximations specifically for Bayesian optimization or other statistical algorithms?

KEY IDEA

The broader motivation is to study a setting where **useful approximations can be constructed for downstream inference using functions only defined through local and global properties rather than an explicit functional expression.**

Status: Computational experiment / preliminary result

Code: <https://github.com/algorithmic-andy/smoothvi>

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Full text available upon request